

Evaluating AI-Generated Code with GitHub Copilot

Eduardo Corpeño

Course Prompts

Catching a scope mismatch in GitHub Copilot's code

Prompt 1:

```
Build a Python program that plays tic-tac-toe.
```

Prompt 2:

```
I was expecting to play against the computer.
```

Prompt 3:

```
Please update the code so the computer plays as 0.
```

Prompt 4:

```
Please modify the source file.
```

Finding the edge case Copilot missed

Prompt 1:

```
Write a Python program that keeps a to-do list saved in a JSON file. It should let the user view their tasks, add a new task, and remove a task by its ID. Each new task should get an ID one higher than the current highest ID in the file.
```

Prompt 2:

```
This program fails when the EOF character is entered. Please avoid a crash in that case and just treat it as an invalid input.
```

The hidden assumption in GitHub Copilot's suggestion

Prompt 1:

```
Write a Python function that takes a list of numbers and removes all the even numbers from it in-place.
```

Challenge: Find and fix the problem in GitHub Codespaces

Prompt 1:

```
Write a Python program that splits a restaurant bill among a group of friends, including tax and tip.
```

Hard-coded secrets in GitHub Copilot's output

Prompt 1:

```
Write a self-contained Python script that authenticates to the GitHub REST API using a username and personal access token. Send a request to https://api.github.com/user, display the authenticated username, and handle authentication and network errors. Use only Python's standard library.
```

Prompt 2:

```
This code is not secure as it contains hardcoded credentials. Please make it right.
```

When GitHub Copilot calls a made-up method

Prompt 1:

```
Create a standalone Python expense ledger using argparse and sqlite3. Use built-in demo expenses when no arguments are provided. Support custom expenses in DESCRIPTION:AMOUNT format and an optional --save DATABASE argument.
```

The GitHub Copilot test suite that missed a bug

Prompt 1:

```
Write a small Python program for an online shop that calculates a customer's shipping quote. Orders totaling $50 or more receive free shipping; smaller orders pay a $6.99 standard shipping charge. Include a function that accepts items as name-and-price pairs and returns the subtotal, shipping charge, and final total. Use only the Python standard library.
```

Prompt 2:

```
Also include a unittest suite covering orders of different amounts and a final-total calculation to expose any faults in the function.
```